

The CoQuí PAW/USPP D -matrix for self-consistent HF/GW: canonical conventions, derivations, and units ledger

CoQuí PAW workstream (plan E1 deliverable)

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Scope. This is the canonical statement of the non-local D -matrix architecture used by CoQuí for ultrasoft (USPP) and PAW pseudopotentials in a Hartree–Fock / self-consistent-GW framework: what is frozen, what is density-dependent, where exchange lives, the normalization and unit conventions, and the derivations behind each choice. It supersedes the scattered statements in the working notes; the corrective plan (`notes/paw_dmatrix_cleanup_plan.tex`) remains the project history, and `dmatrix_paw_generalization.tex` keeps its symmetry-rotation scope. The on-disk converter contract lives in `notes/converter_h5_contract.md`.

Setting. Augmented species carry AE/PS partial-wave pairs $\phi_I, \tilde{\phi}_I$ (stored in $u = rR$ form), projectors β_I with $\langle \beta_I | \tilde{\phi}_J \rangle = \delta_{IJ}$, and the L -decomposed compensation functions $Q_{ij}^L(r)$ (`qfunc1`, r^2 -weighted). I, J run over projector channels ($nh = \sum_b (2l_b+1)$ per species) on atom a ; $n_{s,\mathbf{k},ab}$ is CoQuí’s self-consistent density matrix in the band basis (never `mf.occ()` in production paths). Hartree atomic units ($e^2 = 1$) everywhere in memory; see the units ledger (§7) for the on-disk story.

1 The partition

$$D_a(I, J) = \underbrace{D_a^{\text{stat}}(I, J)}_{\text{frozen, built once}} + \underbrace{D_a^H(I, J)[n]}_{\text{Hartree only, rebuilt from the current } n}, \quad \text{exchange is never in } D. \quad (1)$$

This partition is what makes the operator HF/GW-compatible by construction: the static piece contains no DFT XC and no generator-density screening beyond the frozen core; every density-dependent Coulomb piece is rebuilt from CoQuí’s own n ; exact exchange enters only through Σ_x/K built from ERIs or the direct v_x routine (valence–valence), and through the frozen linear core–valence term inside D^{stat} .

2 I1 — the one-center density matrix (becsum) and its normalization

A *single* becsum definition serves all consumers (Hartree, deeq, exchange, energies):

$$\text{becsum}_a(I, J) = \frac{n_s}{N_k} \sum_s \sum_{\mathbf{k} \in \text{full BZ}} \sum_{ab} P_{a,aI}^{\mathbf{k}*} n_{s,\mathbf{k},ab} P_{b,aJ}^{\mathbf{k}}, \quad n_s = \frac{2}{n_{\text{spin}}} \quad (n_{\text{pol}}=1), \quad (2)$$

with $P_{n,aI}^{\mathbf{k}} = \langle \beta_{aI} | \tilde{\psi}_{n\mathbf{k}} \rangle$.

Normalization derivation. The valence density is

$$n(\mathbf{r}) = \frac{n_s}{N_k} \sum_s \sum_{\mathbf{k}} \sum_{ab}^{\text{BZ}} n_{s,\mathbf{k},ab} \tilde{\psi}_{a\mathbf{k}}^*(\mathbf{r}) \tilde{\psi}_{b\mathbf{k}}(\mathbf{r}), \quad (3)$$

because CoQuí stores per-spin-channel occupations (`occ` $\in [0, 1]$; converter contract §1.5) and equal k -weights $w_k = 1/N_k$ on the full BZ; the spin degeneracy n_s is applied by the *consumer* (`ns_scl` in `v_h/becsum`), never baked into `occ` on disk. The augmentation (compensation) density is by definition $\rho_{\text{aug}}(\mathbf{r}) = \sum_{a,IJ} \text{becsum}_a(I, J) \hat{Q}_{a,IJ}(\mathbf{r})$; substituting the PAW expansion $\tilde{\psi} = \sum_I \tilde{\phi}_I \langle \beta_I | \tilde{\psi} \rangle$ inside the augmentation spheres forces exactly the prefactor of Eq. (2) — the same n_s/N_k that the smooth density carries. Any consumer that re-weights (e.g. the pre-A3 direct-path $w_k = 1/N_k^{\text{IBZ}}$ on a symmetry-reduced mesh) silently breaks the smooth/augmentation consistency of V_H .

Symmetry. When $N_k^{\text{IBZ}} < N_k$ the projections are lifted to the full BZ by the atom-permutation + Wigner- D + Bloch-phase machinery (View-2: full-BZ states are exactly rotated IBZ states in G -space; at time-reversal points the band matrix is complex-conjugated). The lift is cached on the `pseudopot` (`Pskna_full_bz`). Hermitian pair symmetrization $\frac{1}{2}(b_{IJ} + b_{JI}^*)$ stores the real part exactly (the anti-Hermitian imaginary part is inert against symmetric real Q /radial kernels); the anti-Hermitian residual is hard-checked $\leq 10^{-8}$.

3 I2 — the static D : frozen ionic D^0 plus frozen core–valence exchange

$$D_a^{\text{stat}}(I, J) = D_{t(a)}^0(I, J) + \alpha_x E_{t(a)}^{cv}(I, J), \quad \text{built once at pseudopot init, never mutated.} \quad (4)$$

3.1 D^0 : content and backend provenance

$$D_{IJ}^0 = (T^{\text{AE}} - T^{\text{PS}})_{IJ} + \langle \phi_I \phi_J | v_H[n_{Zc}] \rangle - \langle \tilde{\phi}_I \tilde{\phi}_J + \hat{Q}_{IJ} | v_H[\tilde{n}_{Zc}] \rangle. \quad (5)$$

AE–PS kinetic difference plus the frozen-core–screened ionic Hartree terms, including the compensation-moment piece on the PS side. XC-free by construction: no generator V_{xc} , no valence screening.

- **QE:** the UPF dion *is* Eq. (5) by generator convention; `pw2coqui` copies it verbatim ($\div e^2$ at schema ≥ 2). CoQuí adds nothing at read.
- **ABINIT:** the PAW-XML stores only `kinetic_energy_differences`; the converter *assembles* Eq. (5) (`assemble_dij0`): $v_H[n_{Zc}] = v_H[n_{\text{core}}] - Z/r$ and $v_H[\tilde{n}_{Zc}]$ reconstructed from

`zero_potential` + Poisson of $\tilde{n}_{\text{core}} + g_0(q_{\text{core}} - Z)$, same- l pairs, trapezoid to `kkbeta`, deliberately *excluding* ABINIT’s frozen atomic V_{xc1} (ABINIT’s stored v_{loc} is `vbare`+ $v_H - v_{xc1}$ for its DFT SCF; a GW/HF frozen D^0 must not carry it). Validated ≈ 4 digits against ABINIT’s own dumped D^0 .

3.2 Why $\int V_{\text{loc}} \hat{Q}$ is *not* in D^0 (the descreening settlement)

Eq. (5) carries $-\langle \hat{Q}_{IJ} | v_H [\tilde{n}_{Zc}] \rangle$: the *one-center, atomic* reference of the compensation–local-potential coupling, subtracted at dataset generation (standard USPP descreening). In the solid, the full periodic integral must be re-added:

$$H_0 = T + V_{\text{loc}} + \sum_{a,IJ} |\beta_{aI}\rangle \left[D_a^{\text{stat}}(I, J) + \int V_{\text{loc}} \hat{Q}_{a,IJ} \right] \langle \beta_{aJ} |. \quad (6)$$

The $\int V_{\text{loc}} \hat{Q}$ term is the frozen one-body *electrostatic* coupling of the compensation charge to the local potential — neither exchange nor correlation — and is ALWAYS included (`static_h0_D()`, lazily cached). Corroboration: the density path, which adds $\int (V_{\text{loc}} + V_H) \hat{Q}$ on top of `dion`, reproduces QE’s `dee` to $\sim 10^{-6}$ — impossible if `dion` already contained the smooth-grid piece. With this placement the operator identity $H(n) \equiv H_0 + V_H(n)$ holds *exactly* (`h0_plus_hartree_identity`, $\leq 2.5 \times 10^{-13}$).

3.3 E^{cv} : frozen core–valence exact exchange, factor 1

With frozen core orbitals $\{\varphi_c\}$ the core–valence Fock exchange energy is

$$E_x^{cv} = - \sum_v f_v \sum_c \langle \psi_v \varphi_c | r_{12}^{-1} | \varphi_c \psi_v \rangle = \text{Tr}[D_m X^{cv}], \quad (7)$$

i.e. *linear* in the valence density matrix because the core is not varied (Rostgaard-style frozen-core exchange). A linear-in- D_m term is a **one-body operator**: it belongs in H_0 and is captured through $e_{1e} = \text{Tr}[D_m H_0]$ with **factor 1**. The $\frac{1}{2}$ of $\frac{1}{2} \text{Tr}[D_m K]$ applies only to terms *quadratic* in D_m (valence–valence); routing E^{cv} through K would halve it. In the projector representation, expanding ψ_v in AE partial waves inside the sphere and carrying out the closed-shell angular sum,

$$E_{IJ}^{cv} = -\delta_{l_I l_J} \delta_{m_I m_J} \sum_c \sum_L (2l_c + 1) \begin{pmatrix} l_I & L & l_c \\ 0 & 0 & 0 \end{pmatrix}^2 R^L(\phi_I \varphi_c; \varphi_c \phi_J), \quad (8)$$

lm -diagonal, with R^L the Slater radial integral. Validation anchors: hydrogenic $K(1s, 2s) = 16Z/729$, $K(1s, 2p) = 112Z/6561$ ($\sim 3 \times 10^{-11}$); `atompaw <exact_exchange_X_matrix>` for Al-stringent ($\sim 10^{-7}$ rel); factor 1 pinned end-to-end against ABINIT `m_pawdij` (`dijfock_cv`): $\Delta e_{1e}(cv) = -0.5212248$ vs ABINIT -0.521220 Ha ($4.8 \mu\text{Ha}$) on `si222`. Source order at read: `Onecenter/ex_cvij` (ABINIT-XML) $\rightarrow$ native build from `Core/ae_wfc` (`compute_ex_cvij_from_core`, both backends) $\rightarrow$ absent with a loud warning. α_x is the hybrid mixing (1 for HF/GW; the DFT- V_{xc} c–v option remains a declared-but-aborting enum).

4 I3 — the dynamic D : Hartree only, valence only

$$D_a^H(I, J)[n] = \int d^3r v_H[\bar{\rho}[n]](\mathbf{r}) \hat{Q}_{a,IJ}(\mathbf{r}) + \Delta D_a^H(I, J)[\text{becsum}[n]], \quad (9)$$

with $\bar{\rho}[n]$ = smooth valence density + compensation charge, and ΔD^H the radial AE–PS one-center *valence* Hartree (`compute_paw_hartree_atom`, `rad_fac` = 1). For the KS-like operator the $\int V\hat{Q}$ piece uses $V_{\text{loc}} + V_H$ (with the V_{loc} part already frozen into Eq. (6)); for the pure Hartree matrix it uses V_H only. Rebuilt from the *current* n at every evaluation; no QE-density quantity survives anywhere.

Valence-only contract (D2 resolution, 2026-07-25). No frozen-core density enters Eq. (9). The core–valence one-center electrostatics is already inside D^0 (Eq. (5): $v_H[n_{Zc}]$ vs $v_H[\tilde{n}_{Zc}]$ contain the core Hartree). Injecting $\rho_{\text{core}}^{\text{AE/PS}}$ into the dynamic radial Hartree double-counts it. Measured consequence on the ABINIT semicore fixture (whose species export core wavefunctions; every QE fixture has empty core fields and is blind to this): +19.98 Ha excess in the V_H trace, isolated by a term-by-term trace decomposition against an independent numpy probe (smooth-bra 3.7615 ✓, dynamic $\int V\hat{Q}$ 25.4505 ✓, radial one-center 31.77 vs the 11.793 target). Regression guard: `ab_direct_vh_trace_split` pins all four pieces; post-fix direct trace +41.0055 vs target +41.005.

XC. No DFT XC anywhere in D (static or dynamic). The QE `deeq` (which carries $\int V_{xc}\hat{Q}$ and one-center XC) is neither exported by the converters (schema ≥ 1) nor readable by CoQuí. QE-eigenvalue reproduction is a *diagnostic* that assembles $D^{\text{stat}} + D^H[n_{\text{QE}}] + \int V_{xc}\hat{Q}$ explicitly test-side (`dft_eigenvalues`); the PAW residual of that diagnostic is the radial one-center XC of QE’s `ddd_paw`, deliberately unassembled.

5 I4 — exchange: two augmentation modes, one prefactor

Valence–valence exact exchange lives in the direct v_x routine and in the THC/Cholesky ERIs, never in D . Two selectable augmentation-density modes (single source of truth: `vv_compensation` in `paw_exx_options`; `deltaC/K_a` inclusion is mode-derived):

- **moment** (default): moment-restored compensation charge $\hat{Q}$ in the pair densities *plus* the one-center residual $\Delta C = K_{\text{AE}} - K_{\text{PS}}$ contracted with prefactor $s_{\text{oc}} = -1/N_k$. Operator identity: the exact one-center Fock operator — ABINIT `pawdijfock` ($e_{ijkl} + \text{ex_cvij}$) / VASP K_a analogue.
- **shape**: the full AE–PS partial-wave pair density ($\phi_I\phi_J - \tilde{\phi}_I\tilde{\phi}_J$ form factors, `build_grad_tab_full_aeps`) on the dense grid, ΔC *dropped* (it would double count). Operator identity: the Arnaud plane-wave / GW-oscillator operator (ABINIT GW Σ_x , `pawoptosc`).
- In *both* modes core–valence exchange comes from E^{cv} in D^{stat} (§3.3); core–core is the additive constant `exx_core_core`.

The $-1/N_k$ prefactor (derivation summary). The Fock-operator matrix element carries no $\frac{1}{2}$ (it is the functional derivative of the quadratic exchange energy); per k -point sums enter through $w_k = 1/N_k$, and the one-center ΔC contraction is already an energy integral (no volume measure), giving $s_{oc} = -1/N_k$ exactly — *no spin factor, no $\frac{1}{2}$* . Confirmed four independent ways (from-scratch HF derivation; QE `paw_exx.f90` unit-converted; VASP PAW-EXX; THC K_a to machine precision). The historical empirical value $-1/(4N_k)$ was the classic two-wrongs fit: it cancelled a stale fixture whose `deltaC` was $4\times$ too large (pre-/e2² `pw2coqui`). Full derivation: `notes/paw_onecenter_exchange_prefactor.md`.

Fock vs Arnaud, measured. The two modes are different *operators* and must not be conflated when validating against ABINIT: hybrid-DFT Fock (`pawdijfock`) is exact one-center exchange; the GW Σ_x (Arnaud oscillators) is the plane-wave/shape-restored form. On the corrected ABINIT-converted Si mf (post `real_ylm` fix 3956b45): $E_X(\text{shape}) = -1.3175731$ vs converged GW $\Sigma_x -1.31747 \Rightarrow$ **0.10 mHa** agreement; `moment` -1.3178007 (mode split -0.23 mHa on that system; $+3.2$ – 3.4 mHa on the 16-kpt/si222 pair). The earlier “ -47.6 mHa Fock-vs-Arnaud operator difference” is **retracted** — it was the converter’s missing odd- m Condon–Shortley signs, not physics. Hartree always uses moment restoration (compensated smooth + radial one-center is already exact for Hartree — documented, not option-ized).

6 I5–I8 — evaluation semantics, parity, and basis

I5 (route selectability). `nii/nij` vs no-density in `add_vpp_impl` is by design: no density $\Rightarrow$ Eq. (6) only (ERIs supply Hartree/exchange); density passed $\Rightarrow$ the requested dynamic terms are added under separate `add_hartree/add_exchange` flags; `nii` and `nij` overloads produce the identical operator for the same physical density ($\leq 10^{-15}$, `add_vpp.i5_alignment`).

I6 (backend parity). A QE- and an ABINIT-converted h5 for the same system drive the identical CoQuí code path; the only tolerated divergence is dataset *presence* (`ex_cvij`, `Core/`, `exx_core_core`, `madelung_constant`). Known semantic asymmetries (dion provenance, `ae_vloc` content, ΔC quadrature domain) are catalogued in `notes/converter_h5_contract.md` §4. Post-fix converter parity on Si PAW: ΔE_X $87 \mu\text{Ha}$, ΔRPA $23 \mu\text{Ha}$; the residual Δe_{1e} is a benign uniform $v_{\text{loc}}(G=0)$ gauge constant.

I7 (path equivalence). The ERI route (THC/Cholesky) and the direct route are a compatible partitioning of the same Hamiltonian: $J^{\text{ERI}} = J^{\text{direct}}$, $K^{\text{ERI}} = K^{\text{direct}}$ element-wise up to converged ISDF thresholds. Acceptance (Workstream D): V_x route-equivalence STRICT on QE and ABINIT fixtures (`thc_vs_direct.vh_vx`; AB 3.7×10^{-5}), V_H at the THC-truncation scale, Hartree energy THC $\equiv$ ABINIT+one-center AE target to 4.9×10^{-4} Ha. Smooth-only ERIs can never ship silently for augmented species (Cholesky hard-abort, D4).

I8 (AE basis, identity overlap). HF/GW algebra operates in the AE orbital basis $|\psi\rangle = \hat{T}|\tilde{\psi}\rangle$: every matrix element is an AE matrix element represented through smooth orbitals + projector augmentation, and the overlap in all SCF/Dyson algebra is the **identity**. `nii/nij` are AE density matrices; the S -orthonormality of the smooth orbitals is a property of the representation, not an extra metric in the theory.

7 Units and schema ledger

In memory: Hartree everywhere (since plan B4), including `dion`, `ae_vloc/vloc_ps`, ΔC , `ex_cvij`, local potentials. Lengths in Bohr, reciprocal in Bohr⁻¹.

schema_version	energy datasets	/Orbitals@ecutrho	notes
absent	Ry (legacy, QE deeq era)	Ry	readers scale $\times \frac{1}{2}$
1	Ry, deeq-free	Ry	B1
2	Hartree	Ry (QE) / Ha (AB)	B4; readers stop scaling
3	Hartree	Hartree (pinned)	2026-07 audit; dead data dropped

QE-side conversions at write: $/e^2$ on all energy data, $/e^{22}$ on `deltaC` (QE’s `PAW_init_fock_kernel` returns e^{22} -scaled values). ABINIT is natively Ha (no factors). At schema 3, `ecutrho` is the exact inscribed-sphere cutoff of the dense/aug FFT box ($|G|^2 \leq 2e_{\text{cut}}$ reproduces `miller_g`); the pre-pin QE attribute was raw Ry (a $2\times$ inflated sphere on the h5-init path — fixed reader-side for all older files). `occ` is per-spin-channel $\in [0, 1]$; `kpoint_weights` sum to 1. Real harmonics follow QE `ylmr2` everywhere, *including* the odd- m Condon–Shortley signs — channel-diagonal quantities are blind to a sign error there; only off-diagonal $(\mathbf{k}, \mathbf{k}-\mathbf{q})$ pair quantities expose it (the 3956b45 lesson).

8 Validation ledger (anchors)

- $H(n) \equiv H_0 + V_H(n)$: $\leq 2.5 \times 10^{-13}$ (nii/nij $\times$ NCPP/USPP/PAW/PAW-sym).
- becsym symmetry: `diag-nij` $\equiv$ diagonal $\sim 10^{-15}$; nosym reduction exact.
- deeq self-consistency vs QE (historic, density path): $\sim 10^{-6}$.
- `ex_cvij`: hydrogenic 3×10^{-11} ; atompaw X-matrix 10^{-7} ; end-to-end $\Delta e_{1e}(cv)$ $4.8 \mu\text{Ha}$ vs ABINIT.
- One-center prefactor: THC $K_a \equiv$ direct ΔC contraction to machine precision at $s_{\text{oc}} = -1/N_k$.
- Mode identities: $E_X(\text{shape})$ vs ABINIT GW Σ_x 0.10 mHa ; smooth exchange vs ABINIT bare-Coulomb (`fock_icutcoul=3`) μHa .
- Converter parity (Si PAW, post-3956b45): ΔE_X $87 \mu\text{Ha}$, ΔRPA $23 \mu\text{Ha}$, ΔHF $128 \mu\text{Ha}$.
- Route equivalence: V_x strict on QE+AB; synthetic $\ell = 3$ augmentation chain at machine precision (`paw_q_eval_synthetic_l3`).
- AB direct- V_H trace split pinned (`ab_direct_vh_trace_split`); D2 valence-only contract.

Open, tracked, non-blocking: spin-resolved D^H seam (magnetics), SOC through the static assembly (`qq_so/dion_so`), LaNiO₃ $\ell \geq 3$ anomaly follow-up, dense-grid THC memory footprint, psp8-NLCC block convention (`asset-gated`).